

Titanate Catalyst for PET Bottle Resins

DESCRIPTION

Tyzor® ACTivate™ 422 is a hydrolytically stable titanate catalyst specifically designed for PET bottle resin production. Superior ligand chemistry provides excellent catalytic activity, producing PET resin and bottles with superior color and aesthetics.

Tyzor® ACTivate™ 422 benefits can be fine-tuned in a variety of ways to fit each manufacturer's unique requirements for a balance of manufacturing efficiency, product quality and performance improvements in demanding bottle applications.

Application Benefits:

- Improved Process Efficiency and Throughput
- Enhanced Resin and Bottle Properties
- Improved Resin and Bottle Performance

Tyzor® ACTivate™ 422 is a very effective esterification catalyst that can be used to increase the esterification reaction by up to 20%, useful for de-bottlenecking a process. Alternatively the esterification temperature can be lowered to reduce diethylene glycol concentration in the resin.

Tyzor® ACTivate™ 422 is a very effective melt polymerization catalyst that can be used to increase the melt plant rate by up to 15% or to decrease the temperature of the melt polymerization process, resulting in improved product quality. Tyzor® ACTivate™ 422 has been shown to produce bottle resin and bottles with a reduced level of oligomers, acetaldehyde and haze.

Tyzor® ACTivate™ 422 changes the crystallinity properties of the resin compared to an equivalent Sb catalyzed resin showing a higher onset of crystallation on heating and a lower onset of crystallation on cooling. These changes can be exploited by a reduction in IPA content of up to 40%, producing significant financial benefits.

Tyzor® ACTivate™ 422 is hydrolytically stable, which provides exceptional manufacturing versatility. It can be beaded before or during esterification, to either the paste tank or the esterification reactors. Alternatively it can be added to the oligomer line after esterification.

FIGURE 1: Tyzor® ACTivate™ Catalysts Add Value to PET Production and Products

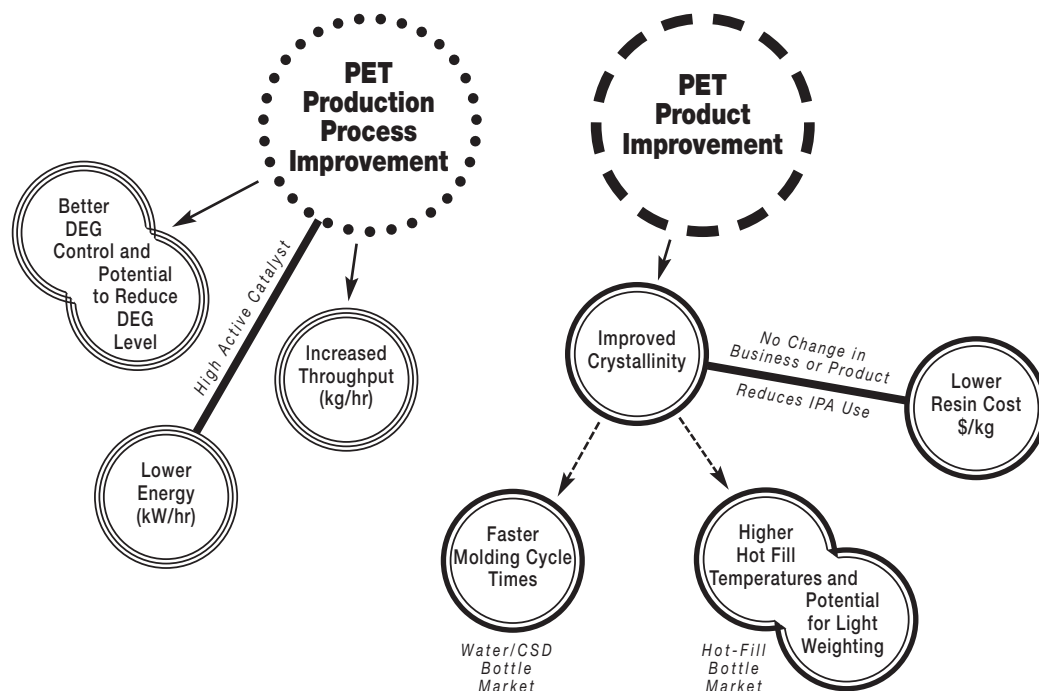


Figure 2 and 3 compare Tyzor® ACTivate™ 422 resin crystallinity performance to antimony resin crystallinity performance.

FIGURE 2: Tyzor® ACTivate™ 422 Resin DSC Trace – Heat-Up

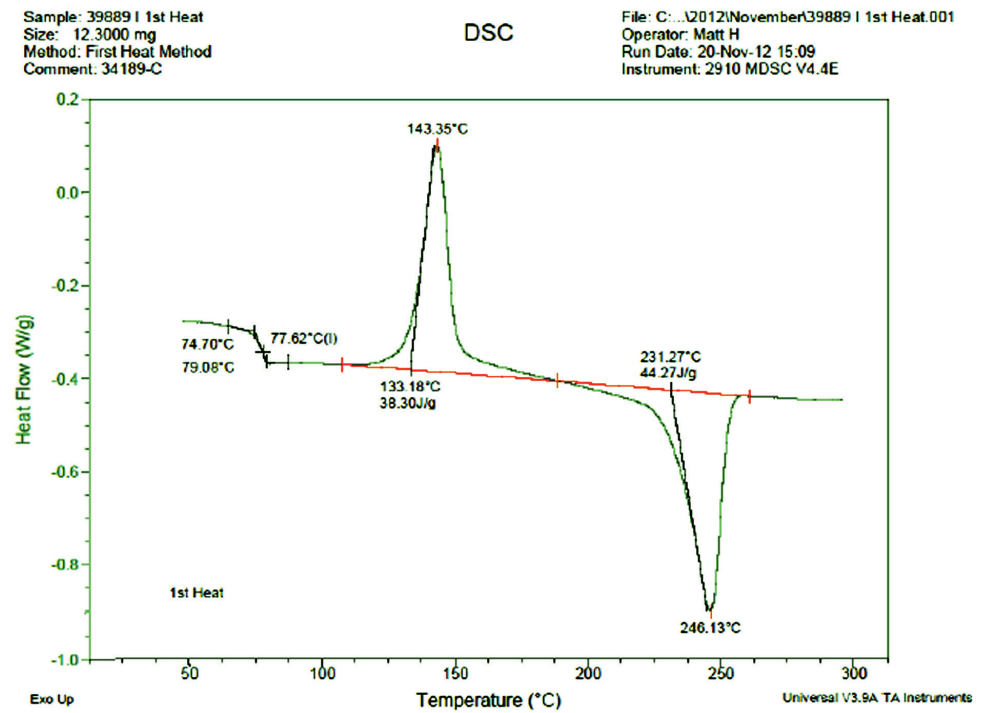


FIGURE 3: Antimony Resin DSC Trace – Heat-Up

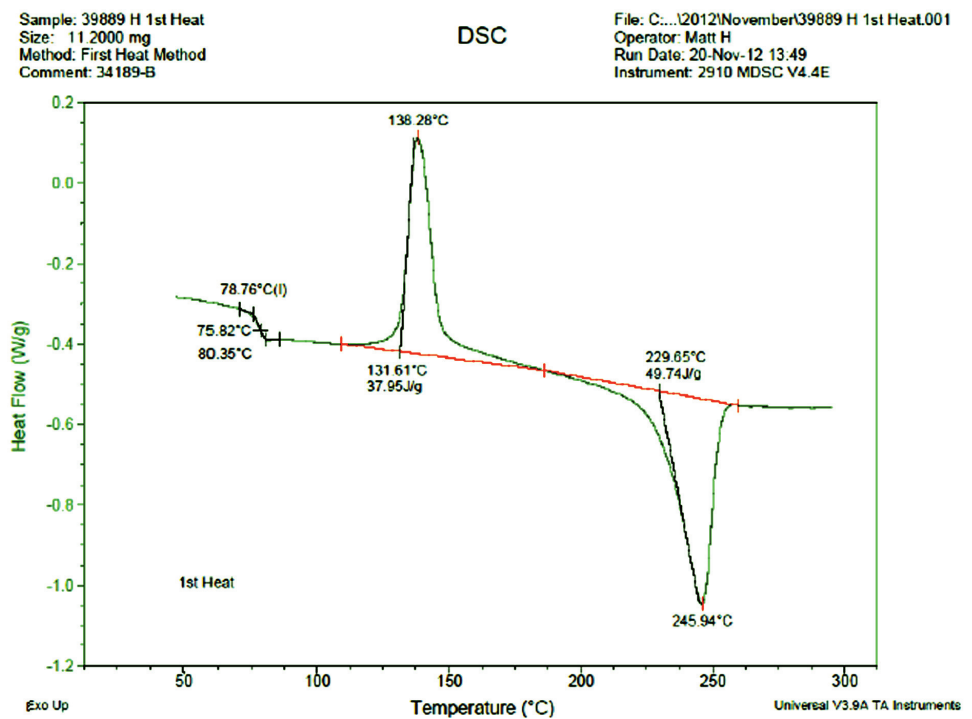


Table 1 provides a rough estimate of the savings that can be realized when using Tyzor® ACTivate™ 422 in various resin applications utilizing different capabilities.

TABLE 1: Cost Savings Comparisons for Dye-Toned Formulations

	ANTIMONY	TYZOR® ACTivate™ 422		
	Standard Formulation	Reduced IPA	Co-monomer Reformulation (Reduced Molding Temperature/Light Weighting/Higher Fill Temperatures)	
Applications	All	CSD/Water	CSD/Water	Hot Fill
Cost Impact PET Producer (\$/MT PET)	0	-3.13	0.75	0.75
Cost Impact Converter (\$/MT PET preforms)	0	0	-22.44	-7.76
Total Cost Impact (\$/MT PET)	0	-3.13	-21.69	-7.01
Additional Product Benefits	0	Reduced Stress Cracking	Reduced Preform AA and Reduced Stress Cracking	Higher Fill Temperatures

Tyzor® ACTivate™ 422 can increase productivity and significantly reduce manufacturing costs without compromising bottle aesthetics by allowing resin manufacturers to reformulate co-monomer levels for improved preform injection molding performance. Injection molding temperatures can be reduced by up to 10 °C (18 °F), reducing cycle time by as much as 8%.

Tyzor® ACTivate™ 422 also can produce bottle resins for hot-fill with a lower level of shrinkage. This can be exploited with higher filling temperatures and/or bottle light weighting, reducing raw material costs and increasing productivity.

Tyzor® ACTivate™ 422 produces PET bottle resins that meet or exceed the properties of antimony-catalyzed PET bottle resins at lower cost. Figures 3 through 5 compare polymer chip and bottle color properties of Tyzor® ACTivate™ 422 resin with an equivalent antimony resin.

FIGURE 3: Color Standards Achieved - Base Chip Color

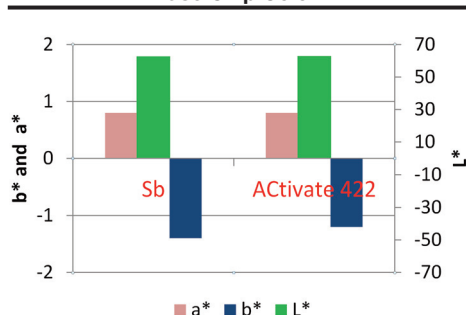


FIGURE 4: Color Standards Achieved - SSP Chip Color

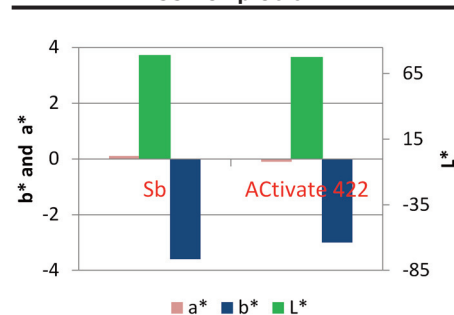
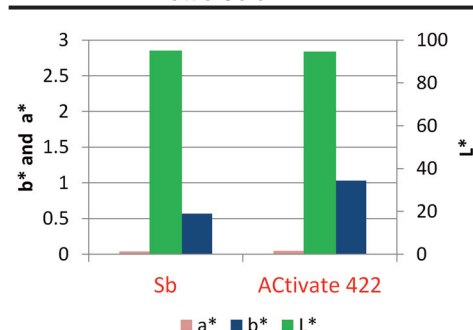


FIGURE 5: Color Standards Achieved - Bottle Color



TYPICAL PROPERTIES
OF TYZOR® ACTIVATE™ 422

PROPERTY	TYPICAL VALUE
Flash point, °C (°F)	> 70 (158)
Specific gravity @ 25 °C (77 °F)	1.14 - 1.18
Viscosity, cPs @ 25 °C (77 °F)	1 to 20
pH	3
Solubility	Miscible in water. Soluble in glycols and alcohols. Insoluble in hydrocarbons.



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